

Size, Power And Multiple Constellations Driving GNSS SOLUTIONS



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In your smartphone you have a miniature GPS receiver, which has a long story behind it. When the first tracking system became operational in 1995, it was as big as a tablet. Today, we have it integrated into our mobile phones in the form of an integrated circuit. The size of a Global Navigation Satellite System (GNSS) chip is getting smaller with every new launch. It has gone down to 25mm².

Reduction in chip size has enabled its use with other handheld devices. This article takes a look at some interesting technologies and improvements driving modern GNSS chips, modules and other solutions.

Navigation is a necessity, with uses ranging from a simple human guidance system to a missile tracking system. Majority of such systems involve a GNSS system, that is, GPS, which is the US guidance system set up with the idea of guiding various military operations. But this century has seen the emergence of several other guidance systems, which also together with GPS have helped improve geolocation.

An era of change

As technology improved over the years, performance of GNSS systems has also improved steadily through availability of multiple constellations of satellites to sync with. Several constellations have

gone live and are giving GPS a run for its money. According to Sudhir N.S., senior manager - GNSS technology, Accord Software, "From the usage point of view, people are still slow in taking up multi-constellation systems. GPS is still hogging most of the number of chips sold, but other systems are catching up."

Globalnaya Navigazionnaya Sputnikovaya Sistema (GLONASS) from Russia, BieDeu from China, GPS Aided GEO Augmented Navigation (GAGAN) from India and Galileo from Europe are some such systems that have improved the tracking. At any given moment we have about 84 operational satellites, solely for the purpose of navigation.

According to Ashish Gulati, country manager, Telit Wireless Solutions, "This means that we are not dependent on the US GPS constellation, and now have the option to go ahead and work with GLONASS, Galileo and others."

Sudhir says in agreement, "I feel multi-constellations will help in cases where open-sky visibility is not possible. If we were to take vehicle tracking as an example, consider a case of a vehicle tracking unit inside a vehicle. Here, the antenna would have to be embedded since customers today prefer self-containing units with internal antennae and sufficient battery backup to prevent tampering."

Hidden navigation systems might receive deteriorated signals from satellites, so having multiple constellations will double the number of satellites and better the chances of a good-quality signal being available to users. It could be for anyone, from common people like us to the military stuck in tough terrain.

Rakesh C.R., field application engineer - GNSS Microelectronics and Solutions, Accord Software, states, "In defence application, which is more strategic in nature, people want multi-constellation availability,

Top trends

Chips have undergone many changes over the past couple of decades. Trends have been:

Size. Reduced to just a few millimetres

Power consumption. Reduced to 1W in certain cases

Integration. Addition of microcontrollers along with GNSS chips help in reduction of additional components

Multiple constellations. Support for GPS, GLONASS, BieDou, GALILEO and other augmented systems

Accuracy. As per the latest statistics, accuracy of up to a few metres